

Programming Assignments

CS101 Introduction to Programming II

```
/*
 * Program: Car Simulation (Simple OOP Example)
 * Author: Marcus Birkenkrahe
 * Date: 04/17/2026
 *
 * Description:
 * This program defines a simple Car class that models basic
 * behavior of a car, including speed, fuel, and distance traveled.
 *
 * The program demonstrates:
 * - Class definition with private data members
 * - Constructor initialization using an initializer list
 * - Member functions to manipulate object state
 * - Encapsulation of data and behavior
 *
 * Operations supported:
 * - accelerate: increase speed and consume fuel
 * - brake: decrease speed safely (no negative speed)
 * - drive: increase distance and consume fuel
 * - refuel: increase fuel level
 * - status: display current state of the car
 *
 * Notes:
 * - Fuel consumption is simplified (integer division)
 * - No upper bounds on speed or fuel are enforced
 * - Negative values are clamped to zero where appropriate
 */

#include <iostream>
#include <string>
using namespace std;

// Class representing a simple car
class Car {
private:
    string model;    // Car model name
    int fuel;        // Remaining fuel units
    int speed;       // Current speed
    int distance;    // Distance traveled

public:
    // Constructor: initialize model, fuel, speed; distance starts at 0
    Car(string m, int f, int s)
        : model(m), fuel(f), speed(s), distance(0) {}

    // Display current status of the car
    void status() const {
        cout << model << " Speed=" << speed
              << " Fuel=" << fuel << "km Range="
              << distance << "km." << endl;
    }

    // Increase speed and consume fuel
    void accelerate(int amount) {
        speed += amount;           // increase speed
        fuel -= amount / 2;        // fuel consumption
        if (fuel < 0) fuel = 0;    // prevent negative fuel
    }
};
```

```

}

// Decrease speed safely
void brake(int amount) {
    speed -= amount;
    if (speed < 0) speed = 0; // prevent negative speed
}

// Drive a certain distance and consume fuel
void drive(int km) {
    distance += km;           // increase distance
    fuel -= km / 2;           // fuel consumption
    if (fuel < 0) fuel = 0;    // prevent negative fuel
}

// Add fuel to the car
void refuel(int amount) {
    fuel += amount;
}
};

int main() {
    // Create a Car object
    Car c{"Mustang", 50, 0};

    // Initial status
    c.status();

    // Simulate driving behavior
    c.accelerate(20);
    c.drive(30);
    c.brake(10);
    c.refuel(15);

    // Final status
    c.status();
}

```

Author: Marcus Birkenkrahe
 Created: 2026-04-17 Fri 17:11